7 Backoffice and multi-node integration

ACTIVE | TOTAL: orig ~7d · forecast ~8.5d | PROGRESS: spent ~0d · rem ~8.5d | forecast end 21 Dec 2026

GOAL

Run a representative multi-node SI-01 setup that consumes reference/start-time data, produces derived timing, and exchanges node-scoped timing data with a reproducible backoffice test environment while preserving local operation during outages.

RESULT

- Multiple TimingNodes retain independent state, sequence and storage.
- Node-scoped reference/start-time data can be received and applied explicitly.
- TeamID interpretation and StageTiming use defined reference-data ownership.
- Local registration continues during upstream outage.
- Pending outbound timing work resumes synchronization after recovery.
- Engineering-API and RabbitMQ transports exercise the same semantic upstream contract.

END DEMO

- Start a multi-node SI-01 test composition.
- Load synthetic RaceData and StageStartTimes for the addressed nodes.
- Feed simulated registrations and inspect TeamID/StageTiming results per node.
- Use the SI-02 Backend pane over IF-03 to send RaceData/StageStartTimes and inspect typed outbound backoffice events.
- Force acknowledgement delay/disconnect, continue local registrations, reconnect and synchronize.
- Switch the SI-02 Backend Simulator to RabbitMQ transport and repeat the recovery flow through the hosted broker against SI-01.

DOCUMENTATION

30-UC R-90 | 41-01-SSD R-85 | 32-06-ISD W-55 | 33-06-IDD W-45 | 43-01-SDD-03 W-70 | 60-SVP R-85

DOCUMENTATION / DECISIONS

D01 DONE

Promote IF-06 UpstreamData semantic contract and public JSON reference representation

IF-06 now defines transport-in...

D02 after D01 DONE

Define reference-data, TeamID, StagePassageList and StageRanking ownership/semantics

EventData owns stable profile...

D03 after D01 DONE

Define source-head, acknowledgement, REQ and LogBook synchronization semantics

Synchronization keeps two...

D04 after D01 D03 DONE

Align backoffice transport SDD plus Engineering API connector / hosted RabbitMQ verification topology

EngineeringApiConnector is...

APPLICATION / PRODUCT

A01 DONE

Adopt and qualify the existing multi-node composition baseline

Adopt the existing multi-node...

A02 after D01 D02 V01 DONE

Implement node-scoped reference-data ingress and TeamID interpretation

Implement the Central-to-node...

A03 after A02 D02 DONE

Implement StagePassageList and StageRanking using the promoted reference-data model

Implement StageTiming using...

A04 after D01 D03 V01 NEXT

Implement transport-independent TIMINGDATA SYNC/REQ LogBook synchronization

Implement transport-independen...

A05 after D04 A02 A04 PLANNED

Implement EngineeringApiBackofficeConnector plus SI-02 Backend pane

Align EngineeringApiConnector...

A06 after D04 V02 PLANNED

Implement/qualify SI-01 RabbitMQ connector plus SI-02 RabbitMQ backend-peer transport

Implement/qualify RabbitMQ on...

VERIFICATION / TEST

V01 after A01 — DONE

Requalify multi-node addressing, isolation, persistence and recovery

Requalified on Java main...

V02 after A03 A05 — PLANNED

Verify reference-data, derived timing and synchronization through the Engineering API backoffice connector

Use the SI-02 Backend pane...

V03 after A06 — PLANNED

Verify SI-02 ↔ hosted RabbitMQ ↔ SI-01 routing/recovery end-to-end

Run ST-3 end-to-end as SI-02...

PLANNING CHANGES

- 2026-10-21 — Refine the upstream architecture after implementation feedback. TimingSystem owns one Upstream with one NodeUpstream per TimingNode. Messaging presents one bidirectional UpstreamStream per NodeId and UpstreamGateway routes those streams to 1..N connector implementations. RabbitMQ consumer-queue/publisher asymmetry remains internal to RabbitMqConnector. EngineeringApiConnector uses the existing role=upstream WebSocket bidirectionally; the earlier HTTP-ingress/WebSocket-egress D04 realization is superseded and Java implementation requires migration.
- 2026-10-13 — Complete A03 on Java main revision 006a92e65e07412cb7caf603f853c90c58b86f5e. StageTiming now derives deterministic elapsed results from committed TimingData plus current EventData/ReserveTagMap/StageStartTimes state, with explicit unresolved outcomes and an explicit ZoneId for civil-time rollover. REV/lifecycle records do not become new stage passages. Ranking remains a future query policy because no ranking/tie/grouping rule is currently specified.

PLANNING CHANGES (CONT.)

- 2026-10-13 — Complete A02 on Java main revision 99d1d5839c7f837d19095f3b253708728952cdc1. The UpstreamData base, ordered per-node RACE_DATA sequence, ReserveTagMap and StageStartTimes SET/DEL state, apply-before-ACK semantics, gap-to-REQ behavior, node UpstreamMessagePort and application UpstreamMessageRouter are implemented and qualified by Java verification and runtime characterization.
- 2026-10-13 — Close A01/V01 by requalifying the existing multi-topology baseline on Java main revision 7d87344725d9deb8c91bfda524b35c085b53b401. VC-ST1-007 and VC-ST1-008 both pass in the integrated -Psystem-test run, confirming independent node addressing, sequence isolation, separate LogBooks and restart recovery.
- 2026-10-13 — Complete D04 engineering upstream transport design. EngineeringApiUpstreamConnector reuses IF-03 POST /api/v1/dev/upstream/message and the existing SI-02 event WebSocket with role=upstream; outbound semantic messages are UPSTREAM_DATA events. No second listener port or SI-02 WebSocket lifecycle is added. SI-02 acts as the Central System Simulator and the connector enters the same UpstreamData boundary as RabbitMQ.

PLANNING CHANGES (CONT.)

- 2026-10-13 — Complete the D02/D03 ordered-stream model before A02 implementation. TIMING_DATA keeps the existing per-node LogBook/IF-05 seqNr. Central-to-node RACE_DATA gets its own per-node source seq and carries incremental reserveTagMap/StageStartTimes changes. SI-01 ACKs RaceData only after ordered semantic application; gaps cause bidirectional REQ from the first missing sequence. SYNC advertises a stream's source head. REQ TIMING_DATA fromSeq=1 is the semantic equivalent of the legacy receive-all/package-id 0000 request.
- 2026-10-13 — Complete the D01 IF-06 baseline. Use one bidirectional UpstreamData model with a public/reference JSON representation: TIMING_DATA embeds structured IF-05 records, while PING uses op=REQ/ANS with optional mode=HB and optional pingId. SI-01 sends heartbeat REQ immediately on connect/reconnect and every 30 seconds; manual PING omits mode. No unanswered-PING timeout policy is fixed yet.
- 2026-10-13 — Use one SI-02 Central System Simulator with two transports: IF-03 for deterministic ST-2 and RabbitMQ for hosted ST-3. Both exercise the same semantic upstream contract; deployment-specific protocol mappings remain below the UpstreamData boundary.

PLANNING CHANGES (CONT.)

- 2026-10-13 — Activate Step 7 after Step 6 closeout and adopt the existing multi-node, node-addressing, separate-LogBook and restart-recovery implementation as the baseline instead of rebuilding that topology. Continue with reference-data, synchronization and backoffice integration work from that baseline.